

Challenge (Habanero)

Q1. Solve the inequality $|x - 5| < 2$. What is the solution set?

- (A) $(-\infty, 3)$
 (B) $(-\infty, 7)$
 (C) $(3, 7)$
 (D) $[3, 7]$
 (E) $(-\infty, -3] \cup [7, \infty)$

(A) $(-6, 2)$

(B) $[-6, 2]$

(C) $(-2, 4)$

(D) $[-2, 4]$

(E) $(-\infty, -2] \cup [4, \infty)$

Q2. A recipe for chicken parmesan requires $|2x - 5|$ ounces of breadcrumbs. If x must be between 2 and 4, what is the range of possible breadcrumb amounts?

- (A) $[0, 3]$
 (B) $[1, 5]$
 (C) $[3, 7]$
 (D) $[2, 6]$
 (E) $[4, 8]$

Q3. The height of a building is $|3x + 2|$ feet. If the height must be between 10 and 20 feet, what is the solution set for x ?

- (A) $(-\infty, -2) \cup (2, \infty)$
 (B) $(-2, 2)$
 (C) $(-\infty, -$

$\frac{8}{3}) \cup (\frac{4}{3}, \infty)$

(C) $(-\frac{8}{3}, \frac{4}{3})$

(D) $(-\frac{8}{3}, \frac{4}{3})$

Q4. A carpenter needs to cut a board to $|x - 10|$ inches. If the board must be between 5 and 15 inches, what is the solution set for x ?

- (A) $(-\infty, 5) \cup (15, \infty)$
 (B) $(-\infty, 0) \cup (20, \infty)$
 (C) $[5, 15]$
 (D) $[0, 20]$
 (E) $[10, 15]$

Q5. Graph the solution set of $|x + 2| \leq 4$ on a number line.

Q6. The cost of shipping a package is $|2x - 10|$ dollars. If the cost must be between 5 and 15 dollars, what is the range of possible package weights?

- (A) $[2.5, 7.5]$
 (B) $[5, 10]$
 (C) $[2, 12]$
 (D) $[7.5, 12.5]$
 (E) $[10, 15]$

Q7. A chef needs to mix $|x + 3|$ cups of flour. If the recipe requires between 2 and 6 cups of flour, what is the solution set for x ?

- (A) $(-\infty, -5) \cup (1, \infty)$
 (B) $[-5, 1]$
 (C) $(-\infty, -1) \cup (3, \infty)$
 (D) $[-1, 3]$
 (E) $(-5, 1)$

Q8. Solve the inequality $|3x - 2| > 7$. What is the solution set?

(A) $(-\infty, -$

$\frac{5}{3}) \cup (\frac{9}{3}, \infty)$

(B) $(-\infty, -\frac{5}{3}) \cup (\frac{9}{3}, \infty)$

(C) $(-\infty, -\frac{5}{3}) \cup (\frac{3}{3}, \infty)$

(D) $(-\frac{5}{3}, \frac{9}{3})$

(E) $(-\infty, \frac{3}{3}) \cup (\frac{5}{3}, \infty)$

Open Ended Questions (FRQ)

Q9. Model the situation: A bridge has a height restriction of $|2x - 5|$ feet. If trucks can be at most 12 feet tall, what is the range of possible values for x ?

Q10. A baker is making a cake that requires $|x - 3|$ cups of sugar. If the recipe can use at most 5 cups of sugar, what is the solution set for x ? Graph the solution set on a number line and express it in interval notation.

Chef's Tips

- Tip 1: Ensure students understand the concept of absolute value inequalities.
- Tip 2: Provide examples of real-world applications to help students model situations using absolute value inequalities.
- Tip 3: Encourage students to graph solution sets on a number line and express them in interval notation.